

22521

24225

3 Hours / 70 Marks

Seat No.

--	--	--	--	--	--	--	--

-
- Instructions* – (1) All Questions are *Compulsory*.
(2) Answer each next main Question on a new page.
(3) Illustrate your answer with neat sketches wherever necessary.
(4) Figures to the right indicate full marks.
(5) Assume suitable data, if necessary.
(6) Use of Non-programmable Electronic Pocket Calculator is permissible.
(7) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. **Attempt any FIVE of the following:** **10**
- a) Enlist different types of concurrency control protocols.
 - b) Enlist any four characteristics of XML.
 - c) Define aggregation.
 - d) Enlist any four application of data mining.
 - e) State any four components of Hadoop.
 - f) Give any four features of Hadoop cloudera combination.
 - g) State any two data types used in Mongo DB.

P.T.O.

- 2. Attempt any THREE of the following:** **12**
- a) Explain any two architectural design for parallel database.
 - b) Differentiate between structured and unstructured data. (Write any four points)
 - c) What is the use of find () in Mongo DB? Explain it with suitable example.
 - d) Explain complex data types with suitable example.
- 3. Attempt any THREE of the following:** **12**
- a) State different approaches of machine learning.
 - b) Compare SQL and NoSQL database system. (Write four points)
 - c) With the help of neat diagram explain Hadoop architecture.
 - d) Explain mobile database with neat diagram.
- 4. Attempt any THREE of the following:** **12**
- a) Explain benefits of distributed database system. (Any four)
 - b) Explain basic datatypes and arrays in MongoDB. (Any two each)
 - c) Explain classification and clustering in data mining.
 - d) Write any four features of R-programming.
 - e) Describe the features of BI components. (Any four points)

5. Attempt any TWO of the following:**12**

- a) Consider following input data for your Map Reduce Program 'PLSQL is better than SQL'. Draw Map Reduce Architecture and explain its phases.
- b) Explain structured types and inheritance in SQL.
- c) Explain object and object identity. Write SQL query for following table.

Class: Employee
Name
Age
Salary
Department
Gender
Store
Paint
Update

6. Attempt any TWO of the following:**12**

- a) Explain Two Tier and Three Tier client server model.
 - b) Explain array and multiset types in SQL with example.
 - c) Compare data mining and data warehousing. (Any six points)
-